

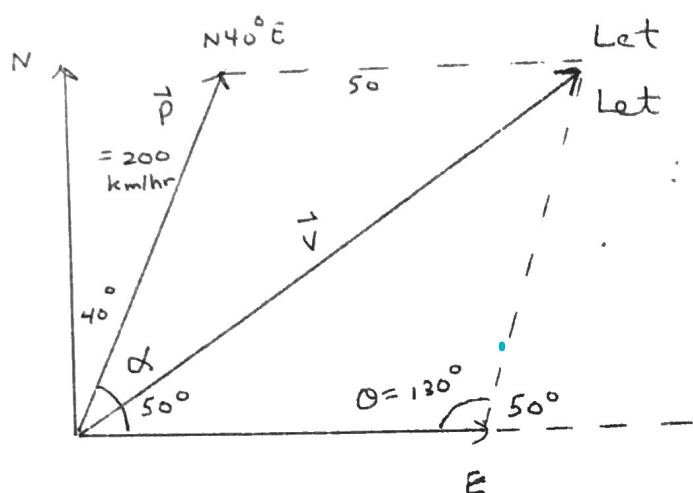
Velocity of a Vector

Example 1

The air speed of a small plane is 200 km/hr steering $N40^\circ E$. The wind speed is 50 km/hr from the west. Find the ground velocity of the plane.

NOTE:

$$\text{Ground Speed} = \text{Air Speed} + \text{Wind Speed.}$$



Let $\vec{p}$ = the velocity of the plane.

Let $\vec{w}$ = " " " " wind.

By Cosine Law: $|\vec{v}|^2 = |\vec{p}|^2 + |\vec{w}|^2 - 2|\vec{p}||\vec{w}|\cos \theta$.

$$|\vec{v}|^2 = 200^2 + 50^2 - 2(200)(50)\cos 130^\circ$$

$$|\vec{v}| = 235.2 \text{ km/hr.}$$

find α to determine the direction:

$$\frac{\sin \alpha}{50} = \frac{\sin 130^\circ}{235.2}$$

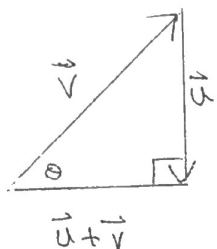
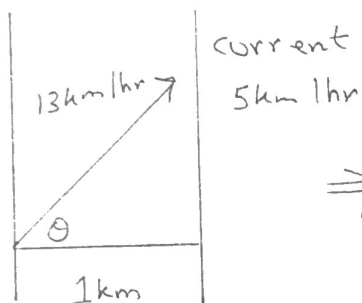
$$\alpha \doteq 9^\circ$$

$\therefore$ The ground velocity is 235.2 km/hr
 $N 49^\circ E$.

Example 2

A canoeist wishes to cross a river 1 km in width. If the current flows at 5 km/hr and the canoeist can paddle at 13 km/hr in still water, determine the following:

- a) the direction the canoeist must head to travel **directly** across the river.



θ = angle canoeist heads

$\vec{u}$ = velocity of current

$\vec{v}$ = " " canoeist

$$\therefore \sin \theta = \frac{5}{13}$$

$\theta \doteq 23^\circ \therefore$ The canoeist must head 23° upstream

- b) the time it will take to across the river.

$$\text{Time} = \frac{\text{Distance}}{\text{Velocity}}$$

$$\text{Time} = \frac{1 \text{ km}}{12 \text{ km/hr.}}$$

$$|\vec{u} + \vec{v}|^2 = |\vec{v}|^2 - |\vec{u}|^2$$

$$|\vec{u} + \vec{v}|^2 = 13^2 - 5^2$$

$$|\vec{u} + \vec{v}|^2 = 144$$

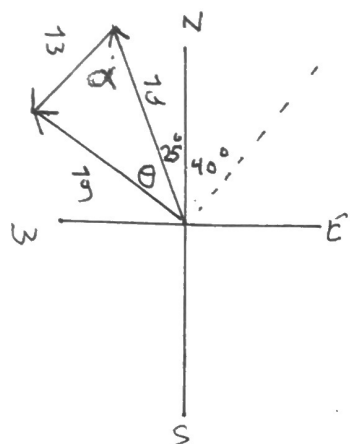
$$\therefore |\vec{u} + \vec{v}| = 12 \text{ km/hr.}$$

$$\text{Time} = \frac{1}{12} \text{ hr.}$$

(OR) 5 minutes.

1. A plane is steering $[N25^\circ W]$ at an airspeed of $325 \frac{km}{hr}$. The wind is from $[N40^\circ E]$ at $90 \frac{km}{hr}$. Find the ground speed of the plane and its course.

Let $\vec{a}$ be the air velocity, $\vec{w}$ be the wind velocity and $\vec{g}$ be the ground velocity.



By Cosine Law:

$$|\vec{g}|^2 = |\vec{a}|^2 + |\vec{w}|^2 - 2|\vec{a}||\vec{w}|\cos(\alpha)$$

$$|\vec{g}|^2 = (325)^2 + (90)^2 - 2(325)(90)\cos(65^\circ)$$

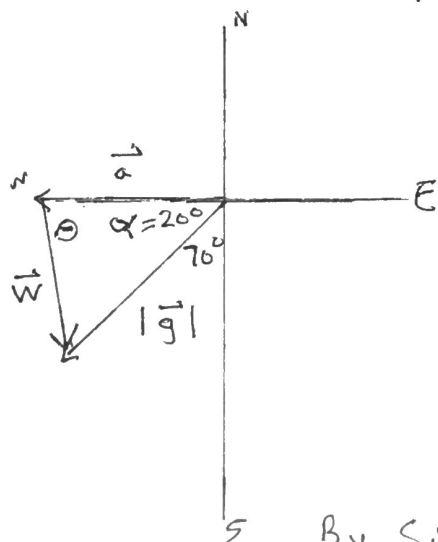
$$|\vec{g}| = 298.33 \text{ km/hr.}$$

$$\frac{\sin(\theta)}{90} = \frac{\sin(65^\circ)}{298.33}$$

$$\theta \doteq 15.9^\circ$$

$\therefore$ Ground speed is 298.33 km/hr on a course of $N40.9^\circ W$

2. A plane is heading $[S70^\circ W]$ with a ground speed of $625 \frac{km}{hr}$. If the pilot is steering west at an airspeed of $665 \frac{km}{hr}$, what must be the wind speed and wind direction?



Let $\vec{a}$ be the air velocity, $\vec{w}$ be the wind velocity and $\vec{g}$ be the ground velocity.

By Cosine Law:

$$|\vec{w}|^2 = |\vec{a}|^2 + |\vec{g}|^2 - 2|\vec{a}||\vec{g}|\cos(\alpha)$$

$$|\vec{w}|^2 = (665)^2 + (625)^2 - 2(665)(625)\cos(20^\circ)$$

$$|\vec{w}| \doteq 227.44 \text{ km/hr.}$$

By Sine Law:

$$\frac{\sin(\theta)}{625} = \frac{\sin(20^\circ)}{227.44} \Rightarrow \theta \doteq 70.0^\circ$$

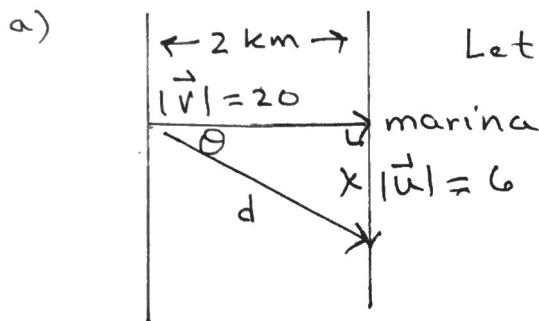
$\therefore$ The wind speed is 227.44 km/hr with direction of $S20^\circ E$.

3. A river is 2 km wide and flows at 6 km/hr. A motor boat that has a speed of 20 km/hr in still water heads out from one bank perpendicular to the current. A marina lies directly across the river on the opposite bank.

- How far downstream from the marina will the boat reach the other bank?
- How long will it take?

Let $|\vec{u}|$ be the velocity of the current.

Let $|\vec{v}|$ be the velocity of the boat.



$$\tan(\theta) = \frac{6}{20}$$

$$\theta \doteq 16.7^\circ$$

$$\tan 16.7^\circ = \frac{x}{2}$$

$$x \doteq 0.6 \text{ km}$$

$\therefore$ The boat will reach the other side
0.6 km downstream from the marina.

b)

$$S = \frac{d}{t}$$

$$t = \frac{d}{S}$$

$$t = \frac{2.09}{20.88}$$

$$t \doteq 0.1 \text{ hours}$$

(OR)

6 minutes

$$* d = \sqrt{2^2 + 0.6^2}$$

$$d \doteq 2.09 \text{ km}$$

$$* S = \sqrt{20^2 + 6^2}$$

$$S \doteq 20.88$$